Penghui Du

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Research Interests

My current work is on precision functional mapping. I am interested in how individual-specific functional network organization can deepen our understanding of human cognition, neurological / psychiatric disorders, and neuromodulation. I am also interested in integration of multimodal neuroimaging data (such as fMRI, PET, EEG and MEG).

Education

2024/08 - 2027/03 **École Polytechnique Fédérale de Lausanne (EPFL)**, *Master in Neuro-X, GPA: 5.40 / 6.*

2020/08 - 2024/07 **Southern University of Science and Technology (SUSTech)**.

- Bachelor in Intelligent Medical Engineering, GPA: 3.84 / 4, Rank 2 / 22.

2023/02 - 2023/06 **University of Zurich (UZH)**, *Visiting Student, Main Subject in Neuroinformatics.*

Research

Harvard University, *Visiting Graduate Student Under Bertarelli Fellowship.*

2026/03 - 2027/03 Supervised by Prof. Randy Buckner. Developed iGENMap, a generative method that learns how individual functional eigenmodes correspond to cortical networks, enabling individual-specific network mapping from substantially less fMRI data than current state-of-the-art approaches require.TODO: add specific time

Investigating precision brain mapping for personalized transcranial magnetic stimulation in depression by comparing empirical, group-level, and individualized targeting strategies and assessing their relative benefits and practical trade-offs.TODO: looks like a review

Max Planck Institute for Human Cognitive and Brain Sciences, *Summer Intern.*

2025/06 - 2025/08 Supervised by Prof. Sofie Valk and Dr. Bin Wan in Cognitive Neurogenetics Lab. Developed a deep learning framework to predict individual brain glucose metabolism from structural and functional MRI features, demonstrating that metabolic organization can be reliably inferred from MRI-based representations and generalizes across subjects and datasets.

Athinoula A. Martinos Center for Biomedical Imaging, *Visiting Undergraduate Student.*

2023/07 - 2024/12 Supervised by Prof. Jingyuan Chen. Investigated the cortical organization of resting-state metabolic connectivity derived from functional PET-FDG. Identified a robust superior-inferior gradient primarily driven by low-frequency dynamics and aligned with known functional and anatomical organization.

Southern University of Science and Technology, *Undergraduate Research Assistant.*

2023/05 - 2023/08 Supervised by Prof. Quanying Liu. We reviewed how large language models can benefit from cognitive sciences, with the goal of integrating cognitive science priors into tests for artificial general intelligence.

2022/09 - 2024/07 Supervised by Prof. Quanying Liu. We explored the effects of distinct drugs on the activity and connectivity of brain networks using zebrafish calcium imaging data.

Selected Publications

2026 / 08 **Du, P.**, Coursey, S., Xu, T., Jamadar, S., Nolin, S., Wan, B., Wey, H., Polimeni, J., Price, J., Liu, Q., & Chen, J. Human Cerebral Cortex Organization Characterized by Functional PET-FDG "Metabolic Connectivity". *Eur J Nucl Med Mol Imaging* (2026). https://doi.org/10.1007/s00259-026-08109-5

2026 / 03 **Du, P.** & Chen, J., Data-Driven Methods in Simultaneous EEG-fPET-fMRI: Opportunities and Challenges, at *2025 59th Asilomar Conference on Signals, Systems, and Computers.* https://doi.org/10.1109/IEEECONF67917.2025.11443781

2024 / 04 Qu, Y., Wei, C., **Du, P.**, Che, W., Zhang, C., Ouyang, W., Bian, Y., Xu, F., Hu, B., Du, K., Wu, H., Liu, J., & Liu, Q. Integration of cognitive tasks into artificial general intelligence test for large models. *iScience*, 2024, 27(4), 109550. https://doi.org/10.1016/j.isci.2024.109550

2024 / 01 Qu, Y., **Du, P.**, Wei, C., Che, W., Zhang, C., Ouyang, W., Bian, Y., Xu, F., Hu, B., Du, K., Wu, H., Liu, J., & Liu, Q. Promoting interactions between cognitive science and large language models. *The Innovation*, 2024, 5(2).

Posters

2026 / 11 **Du, P.**, Du, J., Billot, A., Sun, W., Yeo, B. T. T., Eldaief, M. C., & Buckner, R. L. iGENMap: Individualized Generative Mapping of Cortical Networks Using Functional Eigenmodes. Late-Breaking Abstract, *SfN 2026 Annual Meeting*, Washington, DC.

2026 / 06 Wan, B., **Du, P.**, Ferrari, A., Saberi, A., John, A., Liu, X., Hansen, J., Liu, Z., Misic, B., Kaiser, S., Thompson, P., Warrier, V., Bethlehem, R., Bernhardt, B., Valk, S. & Kirschner, M., EigenBrain: a spatial framework for reconstructing individual brain molecular architecture from multimodal MRI, **Oral Presentation & Merit Abstract Award (6 out of 3240)**

2024 / 06 **Du, P.**, Coursey, S., Xu, T., Polimeni, J., Wey, H., Liu, Q., & Chen, J. Human Cerebral Cortex Organization Estimated by Functional PET-FDG "Metabolic Connectivity", *OHBM 2024 Annual Meeting*.

Awards

2025 / 02 EPFL/HMS **Bertarelli Fellowships Program**

2024 / 07 SUSTech - **Distinguished Graduate Award**

2022 / 10 SUSTech Department of Biomedical Engineering - **Outstanding Student Scholarships**

2022 / 06 China College Student Innovation Training Program – **Undergraduate Research Grant**

2022 / 06 2022 Guangdong Biomedical Engineering Design Competition – **First Prize**

2022 / 05 13th "Challenge Cup" Guangdong Entrepreneurship Plan Competition – **First Prize**

2019 / 05 The 13th Asia and Pacific Informatics Olympiad (APIO) – **Bronze Medal**

2018 / 11 National Olympiad in Informatics in Henan Province (NOIP) – **First Prize**

Talks

2026 / 03 **Human Cerebral Cortex Organization Characterized by Functional PET-FDG "Metabolic Connectivity"**, *at Molecular Connectivity Working Group (MCOS) Talk Series*.

2024 / 02 **The Organization of Human Cerebral Cortex Estimated by Functional PET-FDG: The Promise and Controversy of Metabolic Connectivity** , *at Science on Tap*.

Summer Schools

2023 / 10 **Laminar fMRI Course**, *at Martinos Center for Biomedical Imaging*.

2022 / 08 **Neuromatch Academy Computational Neuroscience Summer School**.

2022 / 07 **CLS-CIBR-IDG Summer School in neuroscience**, *with Merit Student Award*.

Teaching experience

2022 Spring **JAVA Programming A**, *Teaching Assistant*, SUSTech.

2022 Spring **Workshop on Basic Programming**, *Organizer*, SUSTech.

Languages

Chinese: Native speaker.

English: IELTS 7.5.

Programming and Software Skills

Proficient: JAVA, Python, MATLAB, SPM, FSL, LaTeX, Freesurfer, Connectome Workbench.

Working Knowledge: C/C++, AFNI, R, HTML, CSS, Git, Linux, Neurodesk, SimNIBS.